Quantization gains and reduced class agreement after self-organizing map initialization on CIFAR-10

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Abstract

Cooperative prototype initialization improves geometric approximation on CIFAR-10 while reducing class agreement at a small training budget. We compared self-organizing map initialization, winner-only competitive learning, random sampling and greedy k-means++ before identical Lloyd refinement. The paired design comprised 320 models: ten stratified draws, two starts, two training budgets (1,000 and 5,000 images), and two representations (centered RGB and 64 principal components). Development evaluation used 5,000 images. A subsequent frozen evaluation tested every archived model on the official 10,000-image test partition after validating reconstructed training preprocessing against all archived development outcomes. Relative to winner-only learning, self-organizing map initialization reduced test quantization error by 1.3511%, 0.7954% and 0.4174% for 1,000-image RGB, 1,000-image principal components and 5,000-image RGB. These effects survived Holm correction across twelve test comparisons. Adjusted Rand index was significantly lower in both 1,000-image conditions; no normalized mutual information difference survived correction. The test evaluation reproduces the three-condition quantization pattern observed during development and establishes reduced class agreement at the smaller budget. Cooperative initialization therefore changes geometric approximation and semantic partition agreement differently after common refinement.

Keywords: Self-organizing maps, Prototype initialization, Image clustering, Quantization, CIFAR-10

1. Introduction

Representing an image collection with a small set of prototypes raises two practical questions: how closely the prototypes approximate the image features, and how well the resulting groups agree with image categories. Across 320 paired CIFAR-10 models, self-organizing map (SOM) initialization reduced official-test quantization error relative to winner-only initialization after common Lloyd refinement in three of four conditions, by 0.42–1.35%. At the 1,000-image budget, adjusted Rand index (ARI) was significantly lower in both representations. No normalized mutual information (NMI) difference survived Holm correction. These results establish a divergence between geometric approximation and class agreement after cooperative initialization.

A prototype is a representative feature vector, and assigning each image to its nearest prototype partitions a collection into groups. SOM learning arranges prototypes on a lattice and updates the winning prototype together with its neighbors (Kohonen, 1982). De Bodt and colleagues developed the use of SOM learning to initialize competitive vector quantization in 1999 (De Bodt et al., 1999) and examined the approach further in 2004 (De Bodt et al., 2004). This work provides the methodological foundation for testing the effect of neighborhood cooperation after a subsequent refinement stage.

The present question concerns the final image representation obtained after that refinement. Does cooperation during initialization produce a measurable improvement in held-out quantization, and how does the same intervention affect agreement with semantic classes? These outcomes require separate measurement. Harris and de Amorim's extensive initializer comparison evaluates geometric fit and class recovery alongside computational behavior (Harris and de Amorim, 2022). Here, a focused comparison of SOM and winner-only learning connects those evaluation dimensions to one controlled change in the warm start.

CIFAR-10 contains 60,000 color images of ten object classes at a resolution of 32 by 32 pixels (Krizhevsky et al., 2009). We examine this question using 1,000 and 5,000 training images represented either by centered RGB values or by 64 principal components. SOM and winner-only initialization share the starting prototype vectors and image order within each paired execution. Both feed the same batch Lloyd implementation, as do random-sample and greedy k-means++ reference initializers. Evaluation uses a fixed, disjoint 5,000-image development partition followed by the official 10,000-image test partition. Ten training draws and two starts per draw support the paired analysis.

The study contributes estimates of the neighborhood-cooperation effect after refinement, its observed variation across the four budget–representation conditions, and the associated quantization, semantic-agreement and occupancy measurements. Saved splits, fitted prototype banks and analysis records connect the reported effects to the underlying executions. This evidence extends the established SOM initialization

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literature with a reproducible natural-image evaluation focused on the final quality of the learned prototypes.

2. Related work

2.1. Image organization and learned representations

SOM-based image organization has a long experimental history. Laaksonen and colleagues combined maps for visual features with relevance feedback in image retrieval (Laaksonen et al., 2000). Teemu Kinnunen and colleagues studied unsupervised object discovery through self-organization of local image features (Kinnunen et al., 2012). These studies make the input representation and intended use part of the evaluation: the usefulness of a map follows from its behavior on an image task, rather than its visual organization alone.

Contemporary image clustering often learns representations that make semantic grouping easier. Chen and colleagues investigated contrastive visual representation learning in SimCLR (Chen et al., 2020). Van Gansbeke and colleagues separated representation learning from clustering in SCAN (Van Gansbeke et al., 2020). Junnan Li and colleagues combined prototype estimation with contrastive representation optimization (Li et al., 2021). These are complementary strategies for shaping the feature space. Keeping features fixed in the present comparison makes the initializer's effect identifiable within a given space.

2.2. SOM initialization and comparative evaluation

The closest methodological lineage is the SOM–competitive-learning hybrid. De Bodt, Cottrell and Verleysen introduced KSCL, which switches from neighborhood-based learning to winner-only learning, and examined the switching point using one-dimensional probability densities (De Bodt et al., 1999). De Bodt and colleagues subsequently expanded the investigation, examining convergence toward reference quantizers and the evolution of distortion (De Bodt et al., 2004). Their hybrid experiments used online SCL after SOM, and their real-data experiments included two-dimensional lattices (De Bodt et al., 2004). The present experiment measures the effect remaining after a common batch Lloyd refinement on a separate image partition.

The comparison of initialization strategies also has an extensive empirical literature. Harris and de Amorim evaluated 17 initialization variants under shared k-means refinement and considered ARI, inertia and computational performance (Harris and de Amorim, 2022). Their study supplies an evaluation precedent for treating geometric fit and label recovery as distinct outcomes. Our design concentrates on a paired SOM-versus-winner intervention, repeats it across training draws, and evaluates the resulting prototypes in two image representations. Random sampling and greedy k-means++ provide reference initializers with which to interpret its practical magnitude.

Arthur and Vassilvitskii's k-means++ selects centers by distance-sensitive sampling (Arthur and Vassilvitskii, 2007); we evaluate scikit-learn's greedy local-candidate variant (Scikit-learn developers, n.d.). Recent Pattern Recognition Letters studies learn multiple contrastive prototypes (Zhao et al., 2026), hierarchical prototypes (Trivedi et al., 2026), or a weighted consensus graph (Yang et al., 2025). These approaches provide context for prototype quality when representation or consensus construction is learned. Our comparison fixes the features and refinement objective.

2.3. Geometry, semantic agreement and reproducibility

Forest and colleagues jointly optimized an autoencoder and SOM (Forest et al., 2021), and distinguished approximation quality from neighborhood preservation in their metric study (Forest et al., 2020). Johnsson and colleagues examined patch-shared hierarchical SOMs and identified further natural-image and neighborhood evaluation as research directions (Johnsson et al., 2026). These studies motivate measuring map properties and task outcomes separately.

Application-specific evidence reinforces that distinction. Tomi Kinnunen and colleagues evaluated clustering through speaker modeling, cluster validity and processing time (Kinnunen et al., 2011). Ahmed and Soliman studied stability, recovery and interpretation in matched-capacity SOM and k-means comparisons on imbalanced gene-expression data (Ahmed and Soliman, 2026). Here, repeated training draws and paired starts quantify neighborhood cooperation through geometric approximation and external class agreement.

3. Materials and methods

3.1. Dataset and repeated sampling

The experiment used the five official CIFAR-10 training batches. From their 50,000 images, a stratified development partition of 5,000 images, 500 per class, was reserved. The other 45,000 images formed the training pool. Ten class-stratified draws supplied training subsets of 1,000 and 5,000 images, respectively 100 and 500 per class. Within each draw, the smaller subset was nested in the larger. Draws could overlap with one another. Training and development indices were disjoint and were stored in the released split archive.

Two initial starts were used for every draw and condition. Initialization seeds followed 20261107 + 100d + s, where d ranged from zero to nine and s was zero or one. Labels determined stratified sampling and external evaluation. Prototype fitting used numerical features. The factorial design crossed two budgets, two representations, four initializers, ten draws and two starts, yielding 320 executions. Figure 1 shows how the paired comparison and statistical unit relate.

3.2. Image representations

Image intensities were converted to 32-bit floating point and divided by 255. For centered RGB, each image was flattened into 3,072 features. A mean vector fitted to the current training subset was subtracted from both training and development images. For the second representation, 64 principal components were fitted to that centered training subset using randomized singular-value decomposition, with random state 20261107 + d. Development images were transformed by the same fitted

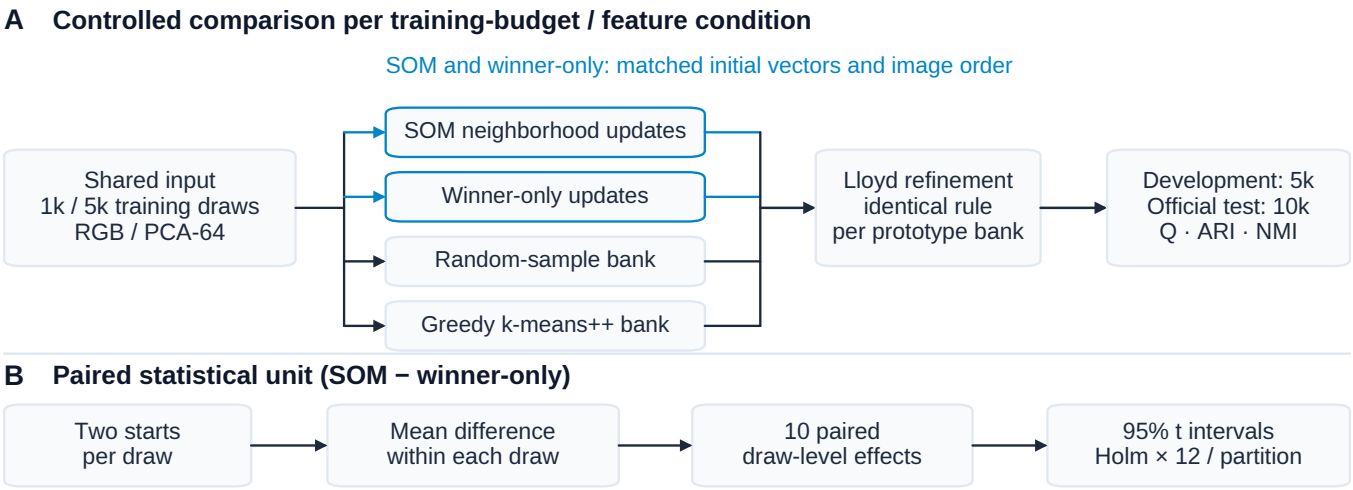


Figure 1: Experimental comparison. Representations and sampled subsets are shared within each condition; SOM and winner-only warm starts additionally share initial vectors and sample order. Every method feeds identical refinement. Two starts are averaged inside each of ten draws before paired inference.

model. PCA fitting was shared across methods and starts in each budget-draw group.

These two feature spaces were analyzed as separate conditions. Every initializer within a condition received the same feature values. The archived environment used Python 3.12.7, NumPy 2.5.4 and scikit-learn 1.9.1; the dependency list accompanies the release. The archived execution platform was macOS on arm64.

3.3. Initialization procedures

Every initializer produced sixteen feature vectors. SOM and winner-only learning began with the same sixteen training images sampled without replacement. Five independently shuffled passes through the training subset were generated from the paired random stream. For each presented feature vector x, the winning prototype b minimized the squared Euclidean distance to x. The learning rate decayed exponentially from 0.5 to 0.05 over all five passes.

The SOM arranged the prototypes on a planar 4 by 4 lattice. A Gaussian neighborhood weight was computed from squared grid distance to the winner, with neighborhood width decaying exponentially from 2.0 to 0.5. The update for prototype j was:

$$w_j(t+1) = w_j(t) + \eta_t h_{jb}(t)[x - w_j(t)]. \tag{1}$$

Here η denotes the learning rate and h the Gaussian weight determined by the positions of prototype j and winner b. Winner-only learning used the same update with weight one at b and zero at every other prototype. Thus, the paired comparison changed the cooperative neighborhood operation while preserving the starting bank and sample order.

Random-sample initialization sent the sampled bank directly to refinement. The greedy k-means++ reference used scikit-learn's kmeans_plusplus routine with sixteen centers and the recorded initialization seed. The routine retained its default local-candidate selection (Scikit-learn developers, n.d.). Logged initialization time measured each method's timed code section; the common sampling of initial vectors and generation of sample order preceded that timer.

3.4. Common refinement and evaluation

Every bank initialized the same Lloyd procedure (Lloyd, 1982). Training images were assigned to their nearest prototype, and each occupied prototype was replaced by the mean of its assigned images. An empty prototype was reseeded using the image with the largest current squared residual; stable ordering and exclusion of already selected reseeding points resolved ties. Refinement stopped when relative Frobenius displacement of the prototype matrix fell below 10^{-6} or after 300 iterations. The displacement denominator included an additive 10^{-12} for numerical stability. Final assignments and quantization were recomputed after the last update.

For N evaluation vectors x_i and final prototype bank W, the quantization score was:

$$Q(W) = \frac{1}{N}\sum_{i=1}^{N} \min_j \|x_i - w_j\|_2^2. \tag{2}$$

The score averages squared Euclidean distance over images; the feature coordinates are summed inside each distance. We use Q to distinguish it from a per-coordinate mean squared error. Semantic alignment was measured by adjusted Rand index (ARI), which corrects pairwise partition agreement for chance (Hubert and Arabie, 1985), and normalized mutual information (NMI) using the library's arithmetic-mean entropy normalization. These measures compared the sixteen learned groups with the ten CIFAR-10 classes. Prototype occupancy was the number of groups receiving at least one evaluation image.

3.5. Paired statistical analysis

The principal comparison was SOM minus winner-only learning. For each condition and metric, differences were averaged over the two starts within each draw. A mean difference,

two-sided paired t-test p-value and 95% t interval were calculated from the ten draw-level differences, with nine degrees of freedom. This preserves the paired design and treats starts as repeated initializations within a draw. The same evaluation images were used for every model within each partition.

Holm's procedure (Holm, 1979) was applied separately within the development and test partitions across twelve comparisons: four budget-representation conditions multiplied by Q, ARI and NMI. The adjusted decision threshold was 0.05. Reported confidence intervals are unadjusted, and the tables identify the adjusted p-values used for decisions. The development family was designated after model execution; the test family and analysis were frozen before test access. Random-sample and greedy k-means++ comparisons were analyzed as secondary, exploratory contrasts. Their confidence intervals and nominal p-values were unadjusted; all twelve greedy k-means++ metric-condition contrasts and all twelve random-sample contrasts are available in the archive. The main text reports the four greedy k-means++ quantization contrasts together. All executions and outcomes were retained.

3.6. Reserved-test evaluation and preprocessing recovery

The official test partition comprised 10,000 images, 1,000 per class. On 7 November 2026, all 320 archived final prototype banks were evaluated once without further fitting or model selection. The evaluation protocol fixed model identities, preprocessing, metrics, numeric validation criteria and the twelve-test Holm family before reconstruction and test decoding. Model training, warm starts and Lloyd refinement were not repeated.

Because the original checkpoints omitted training means and PCA components, preprocessing was reconstructed from the stored training indices. RGB means used the original float32 normalization and in-place centering. Twenty PCA fits used the original randomized solver, 64 components and seeds. Training coordinates used fit_transform; development and test coordinates used transform. Each fit was shared across methods and starts. Validation checked 80 random-initializer banks against their stored sampled coordinates, all 320 training quantization scores, and all 320 development quantization, ARI, NMI and occupancy records. Coordinate errors had to satisfy both an elementwise bound of $2 \times 10^{-4} + 10^{-5}|x|$ and a relative Frobenius bound of 10^{-5}. Quantization errors had to be at most $2 \times 10^{-4} + 2 \times 10^{-6}|Q|$; ARI and NMI errors at most 10^{-12}; occupancy and identities had to match exactly.

A Linux reconstruction failed these fixed criteria and stopped before test decoding. Recovery in the original Python 3.12.7 macOS arm64 environment, with NumPy 2.5.4, SciPy 1.18.1 and scikit-learn 1.9.1, passed all 80 coordinate banks and all 320 model records. The evaluator then transformed test images with those validated preprocessing objects and scored the immutable prototype banks in 512-image batches. It saved per-image assignments, squared residuals and labels, fitted preprocessing arrays, model-level outcomes and paired analyses. Test labels were used only for external evaluation. Independent recomputation from the saved predictions verified all test metrics and statistical summaries.

3.7. Evidence archive and independent checks

Each execution produced a compressed model artifact, a metadata record and a completion event. The release contains 320 checkpoints and 320 sidecar records, split indices, the frozen experimental settings, outcomes, analysis tables and source code. SHA-256 identities link the files to the released manifest. An independent audit checked the manifest and reconstructed the paired statistical analysis from the run-level results. The test supplement adds the frozen evaluation protocol, validated preprocessing, 320 test prediction files, test outcomes and an independent audit. Source and archive links are provided in the data-availability statement.

4. Results

4.1. Quantization after common refinement

SOM initialization decreased development quantization relative to winner-only learning in three conditions (Table 1). The mean difference was -1.778845 for 1,000-image RGB, -0.840733 for 1,000-image PCA-64, and -0.449998 for 5,000-image RGB. Relative to the mean winner-only score within each condition, these differences represented reductions of 1.4860%, 0.9326% and 0.3878%. Their Holm-adjusted p-values were 0.00000381, 0.00108983 and 0.00019666, respectively.

For 5,000-image PCA-64, the mean difference was +0.094862, corresponding to a 0.1064% increase in Q. Its interval included zero and the adjusted p-value was 0.433193. Figure 2A presents all four effects with their uncertainty. The two RGB conditions favored cooperative initialization; PCA-64 showed different directions at the two budgets.

4.2. Semantic agreement

No development ARI or NMI comparison survived the twelve-test Holm correction (Table 2 and Figure 2B-C). ARI differences were negative in three conditions and slightly positive for 5,000-image RGB. The smallest adjusted ARI p-value was 0.059295. NMI differences had both signs, with adjusted p-values from 0.144641 to 0.712862. The analysis therefore identified a quantization benefit in three conditions and no statistically established improvement in either semantic measure.

The complete descriptive results preserve all four initializers. Mean ARI values ranged from 0.03854 to 0.04137 and mean NMI values from 0.08834 to 0.09563 across the sixteen method-condition combinations. These values describe the observed relationship between low-level feature partitions and the external classes. The supplementary tables provide the full method-by-condition means and variability.

4.3. Prototype utilization and convergence

All SOM runs assigned development images to all sixteen prototypes. Mean winner-only occupancy was 14.55 for 1,000-image RGB, 15.75 for 1,000-image PCA-64, 15.40 for 5,000-image RGB and 16.00 for 5,000-image PCA-64. The occupancy pattern accompanied the three quantization-favorable conditions. The observed association supplies a diagnostic for

Table 1: Development SOM-minus-winner quantization effects. Each comparison uses ten paired draw means, each averaging two starts.

Images	Features	Difference in Q	Unadjusted 95% CI	Holm p
1,000	RGB	-1.778845	[-2.081197, -1.476494]	3.80739e-06
1,000	PCA-64	-0.840733	[-1.132463, -0.549003]	0.00108983
5,000	RGB	-0.449998	[-0.573897, -0.326100]	0.000196655
5,000	PCA-64	+0.094862	[-0.016730, +0.206454]	0.433193

Table 2: Development semantic alignment effects. Differences are SOM minus winner-only. Figure 2 supplies the unadjusted intervals.

Images	Features	ARI difference	Holm p (ARI)	NMI difference	Holm p (NMI)
1,000	RGB	-0.002246	0.059295	+0.001195	0.448604
1,000	PCA-64	-0.001849	0.059295	-0.001043	0.712862
5,000	RGB	+0.000024	0.960377	+0.001074	0.712862
5,000	PCA-64	-0.001034	0.201903	-0.002528	0.144641

how representatives were used; its causal role would require an additional intervention.

Across all four methods, recorded Lloyd iteration counts ranged from four to 110. Every stored convergence trace ended at zero relative displacement. All 320 planned executions were archived, and their stored arrays and outcome records passed integrity checks. These records connect the statistical summaries to the fitted model states.

4.4. Recorded computation

Mean initialization-plus-refinement section time for SOM and winner-only, respectively, was 0.7483 and 0.7022 seconds for 1,000-image RGB; 0.0674 and 0.0482 seconds for 1,000-image PCA-64; 3.7740 and 3.4736 seconds for 5,000-image RGB; and 0.3527 and 0.2508 seconds for 5,000-image PCA-64. These measurements describe the instrumented code sections on the execution machine. The twenty unique PCA fit/transform groups totaled 1.6404 seconds, with each group timer repeated in eight result rows. Grouped accounting preserves that shared computation. The supplementary diagnostic table reports every initializer and refinement component.

4.5. Secondary comparison with greedy k-means++

SOM had lower mean development Q than greedy k-means++ in all four conditions (Table 3). The differences were -0.695205 and -0.332653 for 1,000-image RGB and PCA-64, and -0.110056 and -0.131976 for 5,000-image RGB and PCA-64. These post hoc comparisons place the cooperative warm start alongside a distribution-sensitive seeding method. The table reports unadjusted intervals and nominal p-values, separately from the twelve-test SOM-versus-winner Holm analysis.

4.6. Independent reserved-test outcomes

The test evaluation retained the development quantization pattern (Table 4 and Figure 2). SOM reduced Q by 1.623827, 0.720650 and 0.486660 in 1,000-image RGB, 1,000-image PCA-64 and 5,000-image RGB, corresponding to 1.3511%, 0.7954% and 0.4174%. All three passed the separate twelve-test Holm correction. For 5,000-image PCA-64, Q increased by 0.118280 (0.1318%); the unadjusted interval excluded zero, and the Holm-adjusted p-value was 0.081739.

Test semantic agreement exposed a second effect. At 1,000 images, SOM reduced ARI by 0.003105 in RGB and 0.001853 in PCA-64, with Holm-adjusted p-values of 0.029844 and 0.020044. Their unadjusted 95% intervals were [-0.004916, -0.001295] and [-0.002846, -0.000861]. No NMI contrast survived correction. These measured ARI reductions distinguish the test findings from the nonsignificant semantic contrasts on development data.

All SOM banks occupied sixteen prototypes on the test set. The supplement retains the descriptive outcomes for all four initializers and every secondary contrast.

5. Discussion

The experiment establishes geometric gains together with reduced class agreement at the smaller training budget. Cooperative initialization lowered Q in the same three conditions on development and official-test images. On the test set, the 1,000-image conditions also had significantly lower ARI. Neighborhood updates therefore changed the final partition in ways that improved feature approximation and reduced agreement with CIFAR-10 categories. NMI did not show an adjusted difference, so the semantic finding is specifically supported by ARI.

Prototype utilization offers a concrete interpretation to investigate. SOM used every prototype on development images, whereas winner-only initialization left some prototypes unused in the three conditions with favorable Q differences. Cooperative updates may change which parts of feature space receive representatives before refinement. The saved occupancy and quantization outcomes are consistent with that possibility. Establishing whether utilization mediates the gain would require controlling utilization independently of neighborhood updates.

The representation conditions further sharpen the result. PCA-64 retained directions of training variation in a compact feature vector. At the larger budget, its Q comparison changed sign on both evaluation partitions and did not pass either Holm family. This observation suggests examining the initializer jointly with the representation and sample budget. The

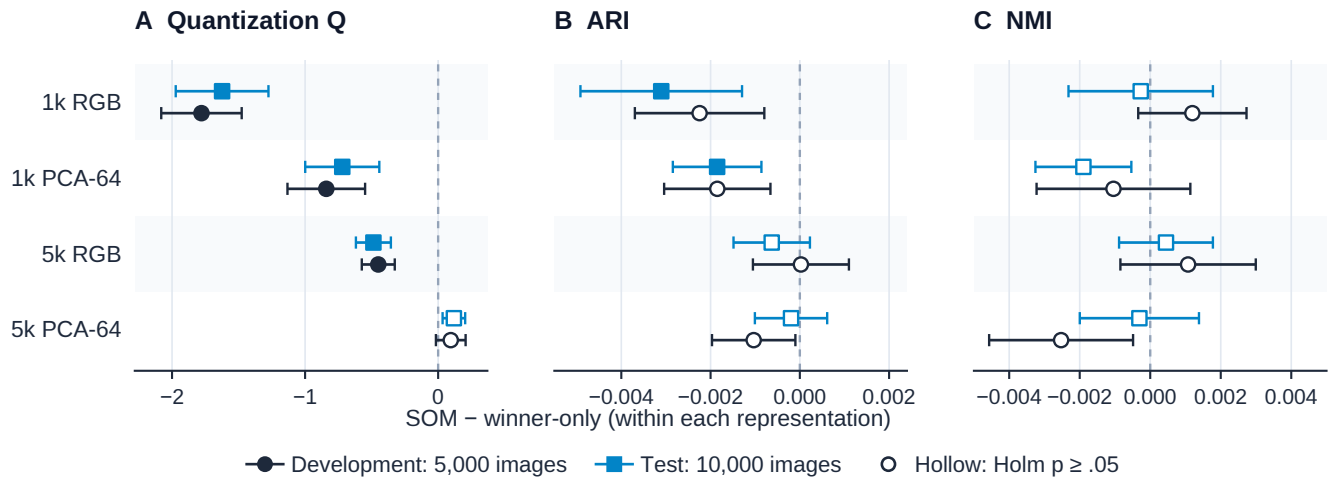


Figure 2: SOM-minus-winner effects on development and reserved-test outcomes. Points average ten paired draw differences, each averaging two starts; bars are unadjusted 95% t intervals. Filled markers pass Holm correction within their partition-specific twelve-test family. Circles show development and squares show test results. Lower Q favors SOM; higher ARI or NMI favors SOM. Training counts are labeled 1k and 5k; PCA denotes PCA-64.

Table 3: Development post hoc SOM-minus-greedy-k-means++ quantization effects. Each estimate uses ten paired draw means; intervals and p-values are unadjusted.

Images	Features	Difference in Q	Unadjusted 95% CI	Nominal p
1,000	RGB	-0.695205	[-1.019959, -0.370451]	0.000918
1,000	PCA-64	-0.332653	[-0.509093, -0.156213]	0.002096
5,000	RGB	-0.110056	[-0.238018, +0.017907]	0.083555
5,000	PCA-64	-0.131976	[-0.253407, -0.010544]	0.036243

condition-level contrasts establish the observed effects within their respective representations and training budgets.

The final-quality measurements extend the SOM-assisted vector-quantization experiments of de Bodt and colleagues (De Bodt et al., 1999, 2004). Their work established the SOM-to-competitive-learning strategy and examined convergence speed and distortion. Here, the endpoints are development and official-test image quality after batch Lloyd refinement, and the measured benefit varies across the four conditions. Joint embedding approaches (Li et al., 2021; Forest et al., 2021) and contrastive prototype learning (Zhao et al., 2026) provide a further application context in which to investigate whether an initialization effect persists as the representation changes.

The computational decision combines outcome and cost. The archive separates initialization and refinement timers and records shared preprocessing. Common sampling and image-order preparation occurred outside those timers. A deployment comparison should charge these operations consistently and assess the measured quantization gains alongside the test ARI reductions.

6. Limitations

Development experiments preceded the frozen 320-run design, and the development Holm family was designated during post-execution analysis. The subsequent test evaluation used a separate family frozen before test access. All models were evaluated on the same official test partition, so the ten paired draws describe variation in training subsets and initialization on a shared evaluation sample. They do not represent ten independent test datasets. The original preprocessing was recovered rather than serialized during training; the validated recovery arrays and endpoint checks are preserved in the test supplement.

The study covers one natural-image benchmark, two low-level representations, one prototype count and one learning schedule. Ten overlapping draws characterize repeated sampling within this fixed training pool, while two starts per draw provide limited information about initialization variability. Nonsignificant semantic comparisons do not establish equivalence. Q values have different feature spaces for RGB and PCA-64 and are interpreted within representation.

The SOM-versus-winner intervention includes both neighborhood structure and additional nonwinner updates. A control that redistributes matched neighborhood weights is needed to attribute an effect specifically to stable geometric adjacency. Direct topology-preservation metrics, a formal mediation analysis of occupancy and fully controlled wall-clock efficiency comparisons are outside the reported analyses. The section timers exclude common bank sampling and sample-order generation, and detailed thread/hardware benchmarking was not recorded. The comparison with the closest predecessors used their primary full texts and targeted method and evaluation pas-

Table 4: Reserved-test SOM-minus-winner quantization effects. Each estimate averages two starts within each of ten draws; Holm correction covers all twelve test contrasts. Q intervals are unadjusted 95% intervals; full semantic intervals are in the supplement.

Images	Features	Difference in Q	Unadjusted 95% CI	Holm p
1,000	RGB	-1.623827	[-1.972273, -1.275382]	2.76234e-05
1,000	PCA-64	-0.720650	[-0.999103, -0.442198]	0.00242368
5,000	RGB	-0.486660	[-0.618108, -0.355212]	0.000168517
5,000	PCA-64	+0.118280	[+0.033424, +0.203135]	0.081739

sages. The literature synthesis is a focused comparison rather than an exhaustive systematic review.

7. Conclusion

Across 320 paired CIFAR-10 models, cooperative SOM initialization reduced official-test quantization relative to winner-only learning in three of four budget-representation conditions after identical Lloyd refinement. The reductions ranged from 0.4174% to 1.3511% and survived a twelve-test Holm correction. ARI decreased significantly in both 1,000-image conditions; no NMI difference survived correction. Together with the development results, the test evidence establishes that cooperative initialization can improve geometric approximation while reducing class agreement at the smaller training budget.

Data and code availability

The study source, 320 checkpoint/metadata pairs, outcomes, splits, statistical tables and integrity manifest are available at https://github.com/Meshaal-Mouawad/2D-Self-Organizing-Map-on-Mini-CIFAR-10/tree/cf741935edc8e3a5da330302dd16a99d86a5f6af/SOM_CIFAR10_Study_2026 . The original CIFAR-10 data are available from their official distribution (Krizhevsky et al., 2009). The reserved-test outcomes, per-image predictions, recovered preprocessing, frozen evaluator and statistical audits accompany this manuscript as supplementary evidence. The linked GitHub commit contains the development release; the later test supplement is supplied with the manuscript package.

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